Session 6: Solutions and facilitation notes

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Exercise numbers follow `Practicals/s6_exercises.md`. All FLUID-ICU numbers come from `Course/Data/key_results.md` (simulated data).

Exercise 1: True or false

#	Answer	One-line explanation
1	**T**	That is the practical reading of a 95% CI: the range of values compatible with the data.
2	**F**	Absence of evidence is not evidence of absence. The CI (−15.2 to 41.6 ml/kg) is wide: the pilot cannot tell.
3	**F**	p is calculated *assuming* the null is true. It is the probability of data at least this extreme under the null, not the probability of the null.
4	**T**	A 95% CI that includes the null value corresponds to p > 0.05 (two-sided), and vice versa.
5	**T**	More patients give more precision, and so a narrower CI.
6	**F**	A tiny p can accompany a trivial effect in a large study. Judge size from the estimate and CI.
7	**T**	This is the strict definition: the 95% refers to the method over repeated studies (the "dance of the CIs").
8	**T**	A small study has low power, so a real effect can be missed (missed fire).
9	**T**	Alpha = type I error rate = risk of a false alarm.
10	**F**	p-values do not "trend". Report the estimate and CI and call it what it is: not statistically significant.
11	**T**	With 20 independent tests and no real effect, the chance of at least one p < 0.05 is about 64% (the jelly-bean problem).
12	**F**	Same patients measured twice: use a **paired** test (paired t-test, or Wilcoxon signed-rank if skewed).

#	Answer	One-line explanation
13	**F**	Switching to one-sided after seeing the data halves the p-value and is not acceptable. Two-sided is the default; a one-sided test needs a strong reason written in the protocol before the data.
14	**F**	Non-inferiority is a different design: a margin fixed in advance, its own hypotheses, sample size and analysis (Session 3).

Exercise 2: Choose the test

#	Test	Why
a	**Independent (unpaired) t-test**	Numerical, roughly symmetric outcome, two independent groups.
b	**Mann–Whitney U**	Numerical but skewed, small groups. (Same choice as Paper 1 for fluid balance.)
c	**Chi-square test**	Binary outcome, two independent groups, large expected counts.
d	**Fisher's exact test**	Binary outcome, very small counts (expected counts < 5).
e	**Paired t-test**	Same patients measured twice; differences roughly symmetric. (Wilcoxon signed-rank if skewed.)
f	**McNemar test**	Binary outcome measured twice in the same patients (paired categorical).

Facilitation: if someone suggests "chi-square" for (f), ask "are these two independent groups?" They are not: the same patients are measured twice, so the data are paired and McNemar applies.

Exercise 3: Match r to the scatter plots

Plot	r	Why
A	**−0.58**	Points fall from left to right, moderate scatter
B	**0.01**	Round cloud, no straight-line trend
C	**0.86**	Tight rising band
D	**0.30**	Weak rising trend, lots of scatter

(The plotted data were simulated with exactly these correlations.)

Bonus: **no**, not on correlation alone. High r means the two methods rise and fall together, not that they give the same value. Look at **agreement**: a Bland–Altman plot (difference against average), the mean difference (bias) and the 95% limits of agreement, and decide whether those differences are clinically acceptable.

Exercise 4: [FLUID-ICU] Interpret the comparison (Q1 together in class; each participant for their chosen question)

Numbers from `Course/Data/key_results.md` (all unadjusted; simulated data).

Q	Test	Result	Model sentence
Q1	Chi-square (binary outcome, two large groups)	134/373 (35.9%) vs 45/245 (18.4%); RD 17.6 points (95% CI 10.7 to 24.4); RR 1.96; p < 0.001	"Twenty-eight-day mortality was 35.9% with a positive and 18.4% with a negative balance (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4)."
Q2	Chi-square	AKI 230/373 (61.7%) vs 71/245 (29.0%); p < 0.001; gradient 29%, 54%, 63%, 75% across balance groups	"AKI occurred in 61.7% of patients with a positive and 29.0% with a negative balance."
Q3	Mann–Whitney (count, piled at 0 and 28)	Median 20 (0 to 28) vs 28 (22 to 28) days; p < 0.001	"Patients with a positive balance had fewer ventilator-free days (median 20 vs 28)."
Q4	Mann–Whitney (lactate skewed)	Median lactate 3.1 (2.2 to 4.5) in non-survivors vs 2.3 (1.6 to 3.4) mmol/L in survivors; mortality < 2: 19%, 2–4: 31%, > 4: 50%	"Non-survivors had higher admission lactate (median 3.1 vs 2.3 mmol/L)."
Q5	Mann–Whitney (skewed, survivors)	Median 7.5 (5.2 to 11.2) vs 4.8 (3.3 to 7.1) days; Hodges–Lehmann difference 2.5 days (95% CI 1.8 to 3.2); p < 0.001	"Among survivors, ICU stay was longer with a positive balance (median 7.5 vs 4.8 days; difference 2.5 days, 95% CI 1.8 to 3.2)." Note: the t-test difference in means (2.7 days) is the wrong summary for this skewed outcome.

Reason for bias (all five questions): confounding by indication. The positive-balance group was sicker at baseline (Table 1).

Exercise 5: Read a regression table (another example, homework)

The table is mock data. The p-values are consistent with the CIs (computed from the log-scale standard error).

1. **Binary outcome** (delirium yes/no), so use logistic regression, which reports odds ratios.
2. **1** (a ratio). An OR of 1 means equal odds.
3. CI excludes 1 (□) or includes 1 (□):

Row	OR (95% CI)	Excludes 1?
Benzodiazepine	2.10 (1.40–3.20)	□

Row	OR (95% CI)	Excludes 1?
Age per 10 y	1.30 (1.12–1.51)	□
Mechanical ventilation	1.60 (0.95–2.70)	□
Female sex	0.85 (0.60–1.20)	□
Dementia	3.40 (1.90–6.10)	□
Sleep protocol	0.62 (0.41–0.94)	□

4. Model sentence: *"After adjusting for age, sex, ventilation, dementia, sleep protocol and illness severity, patients who received a benzodiazepine infusion had about twice the odds of developing delirium (OR 2.10, 95% CI 1.40 to 3.20)."*

5. The OR multiplies for each 10 years: 20 years older gives 1.30 × 1.30 ≈ **1.69 times the odds** (all else equal).

6. **Mechanical ventilation** and **female sex** (CIs include 1).

7. Problems in the conclusion:

- **Causal language** ("cause", "reduces") from an observational cohort: should be "associated with". Confounding by indication is likely (sicker or more agitated patients receive benzodiazepines).
- **"Trend"** for p = 0.08: the CI (0.95–2.70) includes 1, so it is not statistically significant. Report the estimate and CI.
- **"Reduces delirium by 38%"** confuses odds with risk (OR 0.62 is a 38% lower *odds*, not risk; delirium is common, so the risk reduction is smaller) and again implies causation. The sleep protocol was not randomised.

Exercise 6: Read a Kaplan–Meier curve (simulated trial; values computed from the plotted data)

1. About **82%** alive in the restrictive group and about **67%** in the usual-care group at day 28 (read where each curve ends). Accept ±3%.

2. Each tick is a **censored** patient: follow-up stopped (lost to follow-up, transferred, withdrew) without death being observed. They count as alive up to that day.

3. **67** usual-care patients were at risk at day 21. The number falls because of **deaths and censoring**. In the restrictive group 16 died but 33 left the risk set (from 100 to 67 at day 28) because 17 were censored before day 28.

4. The log-rank test gives evidence that the two survival curves differ (p = 0.009: results this different would be unusual if the strategies had the same survival). It does **not** give the size of the effect, its 95% CI, or any adjustment for confounders (and in a trial, it is not "proof" of causation without a good design).

5. **Cox regression** gives the hazard ratio with a 95% CI. Check the **proportional-hazards** assumption: curves that cross or converge strongly suggest the HR is not constant over time.

6. No: without numbers at risk the reader cannot see how many patients support each part of the curve; the right-hand end may rest on very few patients. Reporting guidelines and reviewers expect them.

Exercise 7: Why is it missing? MCAR, MAR or MNAR

#	Mechanism	Consequence for complete-case analysis
a	**MCAR**: an accident unrelated to the patients	Unbiased, just less precise (fewer patients)
b	**MAR**: depends on admission time, which is recorded	Possibly biased if night admissions differ; multiple imputation using admission time can correct it
c	**MNAR**: depends on the missing value itself (the worst pain)	Biased (pain looks better than it is); no guaranteed fix: sensitivity analysis
d	**MNAR** (deteriorating patients are transferred); at best MAR if the reason for transfer is recorded	Biased (mortality underestimated); chase day-28 status by phone/records; sensitivity analysis
e	**MCAR** (missing by design for one month, unrelated to patients), unless patients in the first month differed	Usually unbiased; check whether early patients differed
f	**MAR**: depends on age, recorded for all	Biased towards younger patients' quality of life; multiple imputation including age

Bonus (d): report the main result plus a **best-case** (all transferred patients alive) and **worst-case** (all died) analysis, or vary the assumed mortality among transferred patients. If the conclusion holds under all scenarios, it is robust.

Exercise 8: [FLUID-ICU] Interpret the model (Q1 together in class; each participant for their chosen question)

Numbers from `Course/Data/key_results.md` (simulated data).

Q	Crude	Adjusted (complete case, n = 527 unless stated)	Multiple imputation (from the pack)
Q1	OR 2.49 (1.70 to 3.70), n = 618; same 527 patients 2.22 (1.48 to 3.39)	OR 1.62 (1.02 to 2.59), p = 0.044; Cox HR crude 2.17 (1.55 to 3.05), adjusted 1.46 (1.00 to 2.13), p = 0.053; PH test p = 0.10	MI (m = 20, n = 620): OR 1.70 (1.09 to 2.64), p = 0.020
Q2	OR per 10 ml/kg 1.37 (1.27 to 1.47)	1.27 (1.16 to 1.40), p < 0.001	n/a
Q3	−1.48 (−1.84 to −1.13) days per 10 ml/kg	−0.61 (−1.00 to −0.21) days, p = 0.003	n/a
Q4	OR per 1 mmol/L: CC 1.38 (1.23 to 1.55) vs MI 1.44 (1.28 to 1.61)	Adjusted (age, sex, diabetes, CKD, HF): CC 1.39 (1.24 to 1.57), n = 527 vs MI 1.45 (1.29 to 1.64), n = 620; + SOFA: CC 1.21 (1.05 to 1.39) vs MI 1.23 (1.07 to 1.42)	Lactate missing 93/620 (15.0%), mostly less sick patients (SOFA 5 vs 8; mortality 19.4% vs 30.6%): not MCAR, MAR plausible

Q	Crude	Adjusted (complete case, n = 527 unless stated)	Multiple imputation (from the pack)
Q5	Ratio of geometric means 1.50 (1.34 to 1.68)	1.24 (1.10 to 1.39), p < 0.001 (n = 366 survivors with complete data)	n/a

1. Q1 confounders (pre-specified in the SAP): age, SOFA, lactate, vasopressor, mechanical ventilation, CKD. AKI and dialysis were deliberately NOT adjusted for (they occur after day 3, so they are outcomes or mediators).
2. In every question the adjusted estimate is closer to "no effect" than the crude one: part of the crude association was severity (confounding by indication).
3. All adjusted CIs above exclude the null, but Q1's lower limit is close to 1 (1.02; Cox 1.00): a modest, imprecise association.
4. Complete case 527 of 620; the left-out patients were mostly those without lactate, who were less sick.
5. MI changed the estimates only slightly (Q1 from 1.62 to 1.70; Q4 from 1.39 to 1.45): the conclusions are robust to how missing lactate was handled.
6. (Q1) Day-28 survival 81.6% (negative) vs 64.1% (positive); log-rank p < 0.001 (unadjusted). Adjusted OR 1.62 (p = 0.044) and HR 1.46 (p = 0.053) are in the same direction and of similar size with overlapping CIs; whether p falls just below or above 0.05 does not change the interpretation.

Model sentences (Q1): "Positive fluid balance was associated with higher 28-day mortality (crude OR 2.49, 95% CI 1.70 to 3.70). After adjustment for age, SOFA, lactate, vasopressor use, mechanical ventilation and chronic kidney disease, the association was smaller (adjusted OR 1.62, 95% CI 1.02 to 2.59)."

Exercise 9: Proposal builder 6 (what a good answer looks like)

Sample size (running example): 40% vs 30% mortality, alpha 5% two-sided, power 80% gives 356 per group (spreadsheet sheet 2). Dividing by (1 − 0.10) gives 396 per group, 792 in total.

Analysis plan: describe with mean (SD) / median (IQR) / n (%); compare 28-day mortality with a chi-square test; report RR and risk difference with 95% CIs. Observational designs: adjusted OR (logistic) or HR (Cox) with confounders chosen in advance, about 1 model term per 10 events.

Missing-data plan (good enough): names the variables likely to be missing and why; prevention; main handling (complete case if little missing and plausibly MCAR, otherwise multiple imputation); how missing data will be reported (n missing per variable, who). Level 2 only touches on imputation; imputation and sensitivity analyses in depth are in Level 3, Module H. At Level 1, the focus is on reporting (n missing per variable, flow diagram). FLUID-ICU's SAP section 10 is a model.

Element	Good enough	Needs work
Sample size	4 inputs with a stated source	"All patients in one year"
Analysis plan	Summary + test/model + effect measure with 95% CI	"Data will be analysed with SPSS"
Confounders	Named, justified, within events-per-variable	Stepwise selection; mediators adjusted

Element	Good enough	Needs work
Missing data	Expected gaps + prevention + reporting	Nothing, or "missing data will be excluded"